

2.1: Linear Functions

Definition. (Slope of a Nonvertical Line)

If (x_1, y_1) and (x_2, y_2) are any two distinct points on a nonvertical line L , then the slope m of L is given by

$$m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

Example. Compute the slope of the line passing through the points

$$(x_1, y_1) = (1, 1) \text{ and } (x_2, y_2) = (4, 2)$$

$$m = \frac{2-1}{4-1} = \boxed{\frac{1}{3}}$$

$$(x_1, y_1) = (3, -1) \text{ and } (x_2, y_2) = (-2, 8)$$

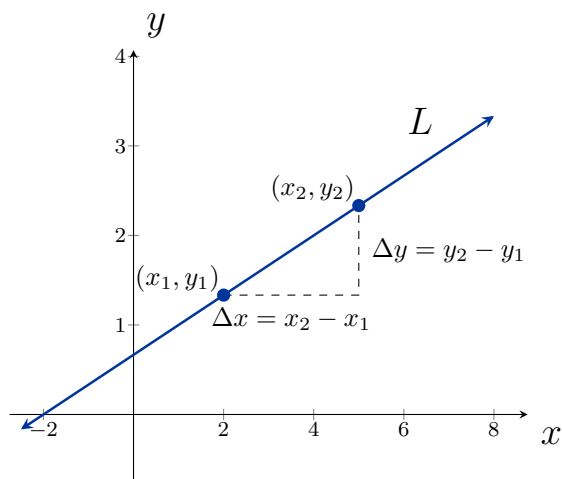
$$m = \frac{8 - (-1)}{-2 - 3} = \boxed{\frac{9}{-5} = -\frac{9}{5} = \frac{-9}{5}}$$

$$(x_1, y_1) = (3, 2) \text{ and } (x_2, y_2) = (-1, 2)$$

$$m = \frac{2-2}{-1-3} = \frac{0}{-4} = \boxed{0} \leftarrow \text{Horizontal line}$$

$$(x_1, y_1) = (4, 1) \text{ and } (x_2, y_2) = (4, 4)$$

$$m = \frac{4-1}{4-4} = \frac{3}{0} \text{ DNE } \leftarrow \text{Vertical line}$$



Definition. (Point-Slope Form of an Equation of a Line)

An equation of the line that has slope m and passes through the point (x_1, y_1) is given by

$$y - y_1 = m(x - x_1)$$

Example. Find the equation of the line going through the points

$$(x_1, y_1) = (-2, 1) \text{ and } (x_2, y_2) = (3, -2)$$

$$m = \frac{-2-1}{3-(-2)} = \frac{-3}{5} \Rightarrow y - (-2) = -\frac{3}{5}(x - 3)$$

$$(x_1, y_1) = (3, 4) \text{ and } (x_2, y_2) = (-1, 4)$$

$$m = \frac{4-4}{-1-3} = \frac{0}{-4} = 0$$

$$y - 4 = 0 \quad \cancel{(x - (-1))}$$

$y = 4$

 ← Horizontal line

$$(x_1, y_1) = (2, 0) \text{ and } (x_2, y_2) = (2, 1)$$

$$m = \frac{1-0}{2-2} = \frac{1}{0} \text{ DNE}$$

$x = 2$

 ← Vertical line

Definition. (Slope-Intercept Form of an Equation of a Line)

An equation of the line that has slope m and intersects the y -axis at the point $(0, b)$ is given by

$$y = mx + b$$

Example. Rewrite the equations in the previous example in slope-intercept form.

$$y - (-2) = -\frac{3}{5} (x - 3)$$

$$y = -\frac{3}{5}x + \frac{9}{5} - 2$$

$$y = -\frac{3}{5}x - \frac{1}{5}$$

Example. Marcus currently has 200 songs in his music collection. Every month, he adds 15 new songs. Write a formula for the number of songs, N , in his collection as a function of time, t , the number of months. How many songs will he own in a year?

t is number of months
 N is number of songs

$$N(t) = 15t + 200$$

$$N(12) = 15(12) + 200 = 380 \text{ songs}$$

Forms of Linear Equations

General form: $Ax + By = C$

Point-slope form: $y - y_1 = m(x - x_1)$

Slope-intercept form: $y = mx + b$

Vertical line: $x = a$

Horizontal line: $y = b$